# Implementing Several Techniques for Data Analytics

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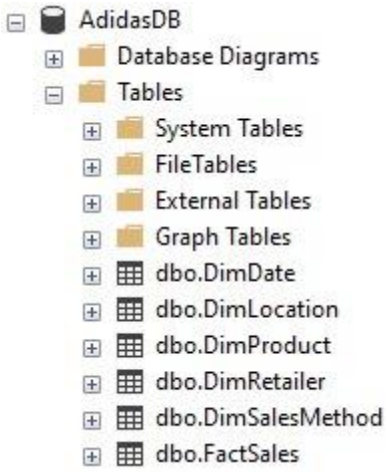
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## PART I – DATA WAREHOUSE / DATA MART




## 1. INTRODUCTION:

The source system used for analysis is the Adidas Sales Dataset. This dataset has been downloaded from Kaggle. It contained a single excel sheet which comprised of Adidas products sales data for a specified timeframe. It contained information such as total sales, number of units sold, price per unit among the total 14 columns to analyze the sales of Adidas products.

Link: https://www.kaggle.com/datasets/heemalichaudhari/adidas-sales-dataset?resource=download

Microsoft SQL Server Management Studio and Visual Studio 2022 were used to create a data warehouse which enabled handling processing of the Adidas sales dataset efficiently. The new system helped to organize and understand data in a better way thus leading to the derivation of valuable insights from the Adidas sales dataset.

### 1.2. REASONS FOR SELECTING THE SUBJECT AREA AND DATA:

The reason behind choosing Adidas Sales Dataset for creation of data warehouse is that it provides detailed information about the product sales data ranging from the product sales, revenue to the operating profit. This dataset contains detailed information that can provide valuable insights to organizations that focus on building effective strategies to analyze product sales trends, identify successful products, and devise strategies for marketing campaigns and future sales.

### 1.3. Vision and Goals:

The vision behind utilizing Adidas sales dataset for the creation of a data warehouse is to help organizations understand the trends in sales and take business decisions that lead to the organization's success. The goal of this project is to provide valuable insights into customer behavior, sales trends, and product performance by performing analysis on the data warehouse created using the Adidas sales dataset. These insights can be helpful to organizations in optimizing their inventory management, refine their pricing strategies and identifying opportunities for their future growth and expansion.

### 1.4. Key Stakeholders:

The key stake holders here are:

Adidas: Where products are manufactured, and orders are placed

Customers: Who influence the sales trend and revenue through their purchases

Retailers: Who sell Adidas products through their stores or online platforms

Supply chain partners: Partners who are involved in the supply chain such as manufacturers, distributors and logistic providers.

Investors and shareholders: Who invest and hold a share in the company

### 1.5. Business requirements:

The business requirements for the Adidas Sales Dataset data warehouse are:

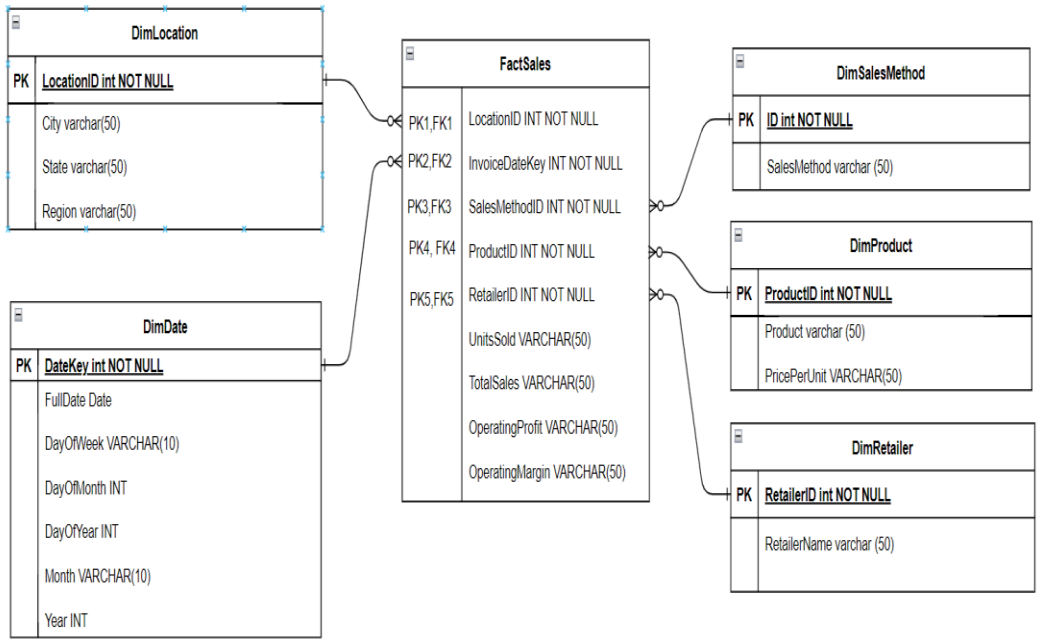
- The data warehouse should display sales performance across different geographical region
- The data warehouse should distinguish sales data based on relevant factors such as product category and sales channels
- The design of the data warehouse should accommodate the expanding dataset seamlessly

It should also enable the analysis of sales trends and patterns over time with the ability to disaggregate data by year, month, or other time periods.

## 2. Schema / Dimensional Model:

Initially I created a star schema to design the data warehouse (Corr & Stagnitto, 2011) as follows: This schema is designed in this way because I have spitted the dataset into Dim and Fact tables. All possible entities are created as Dimension and all data which can be measured is put in Fact tables.

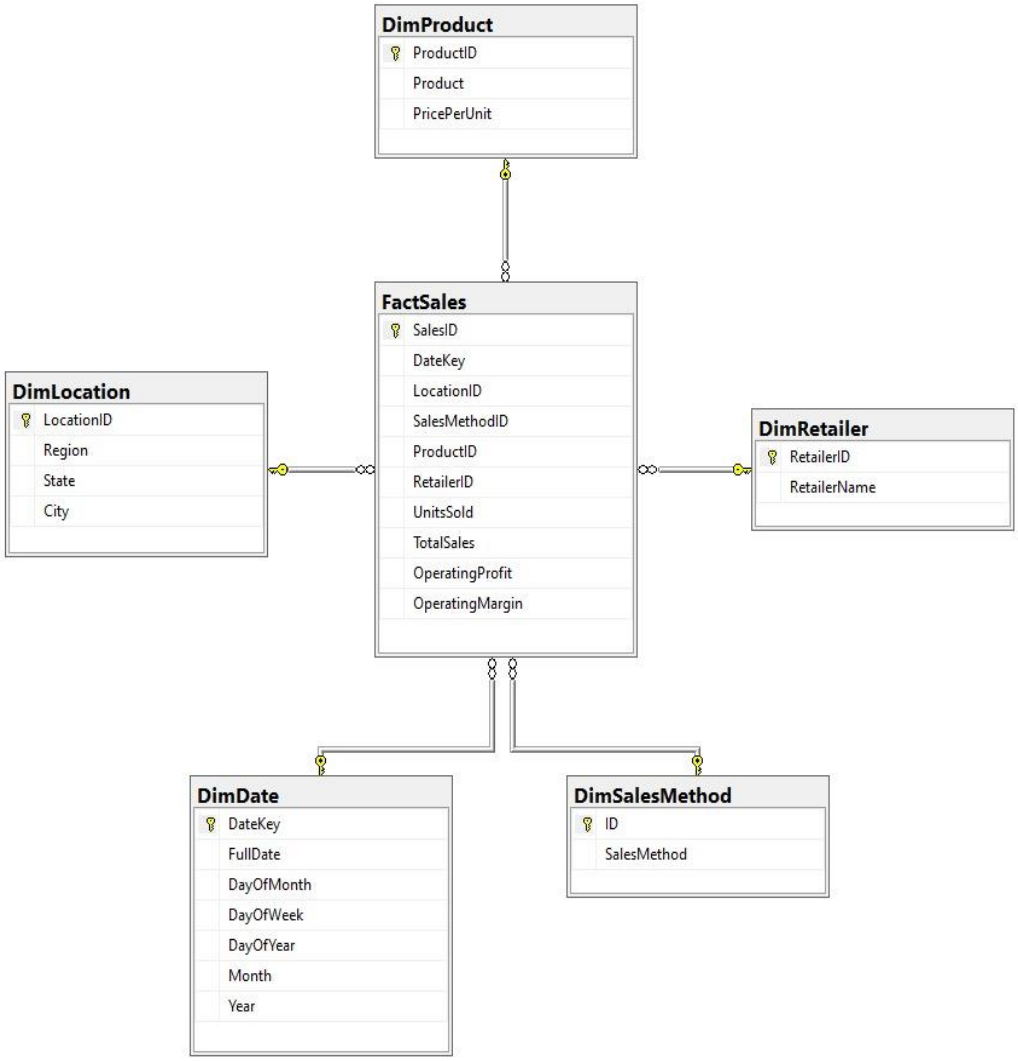
All tables are connected with one other using the Foreign Keys and primary keys are clearly depicted. The design allows to keep data independent and reduce the data redundancy. It also allows data to relate to each other as and when required.




## 3. IMPLEMENTATION OF DATA WAREHOUSE /DATA MART:

**Data Diagram:**

The data diagram is generated and exported using SQL Server Management Studio after creating all the tables and adding relationships between all the tables.



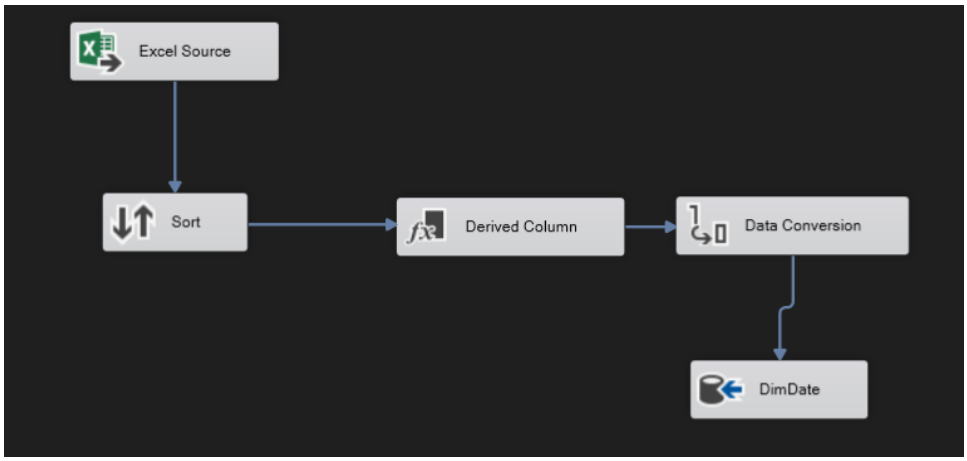

## 4. ETL TO POPULATE THE DATA WAREHOUSE / DATA MART:

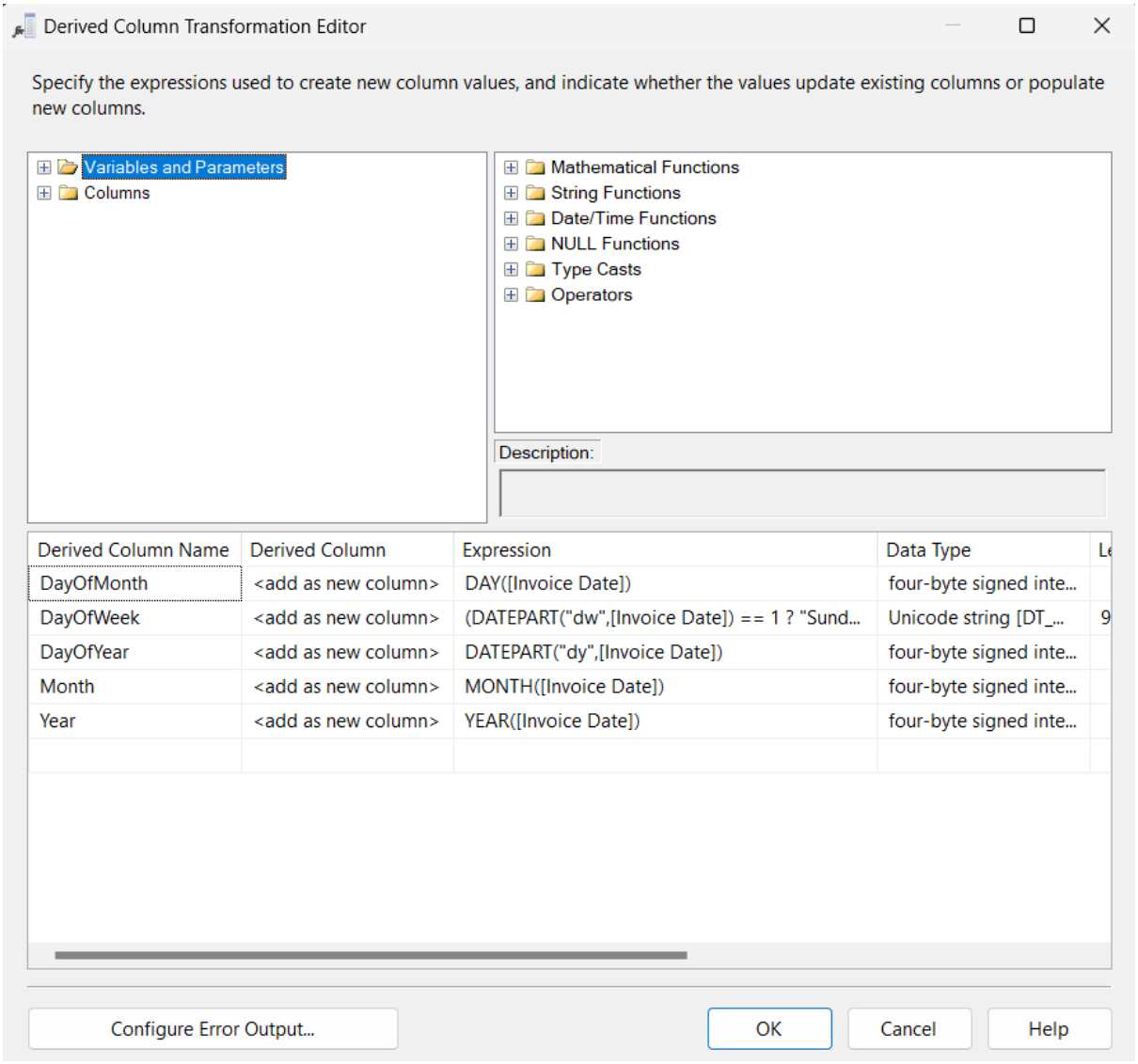
**Creation of the data warehouse using the ETL Procedure:**

SSIS Packages created for the table creation. (Radivojević, Sarka and Durkin, 2019)

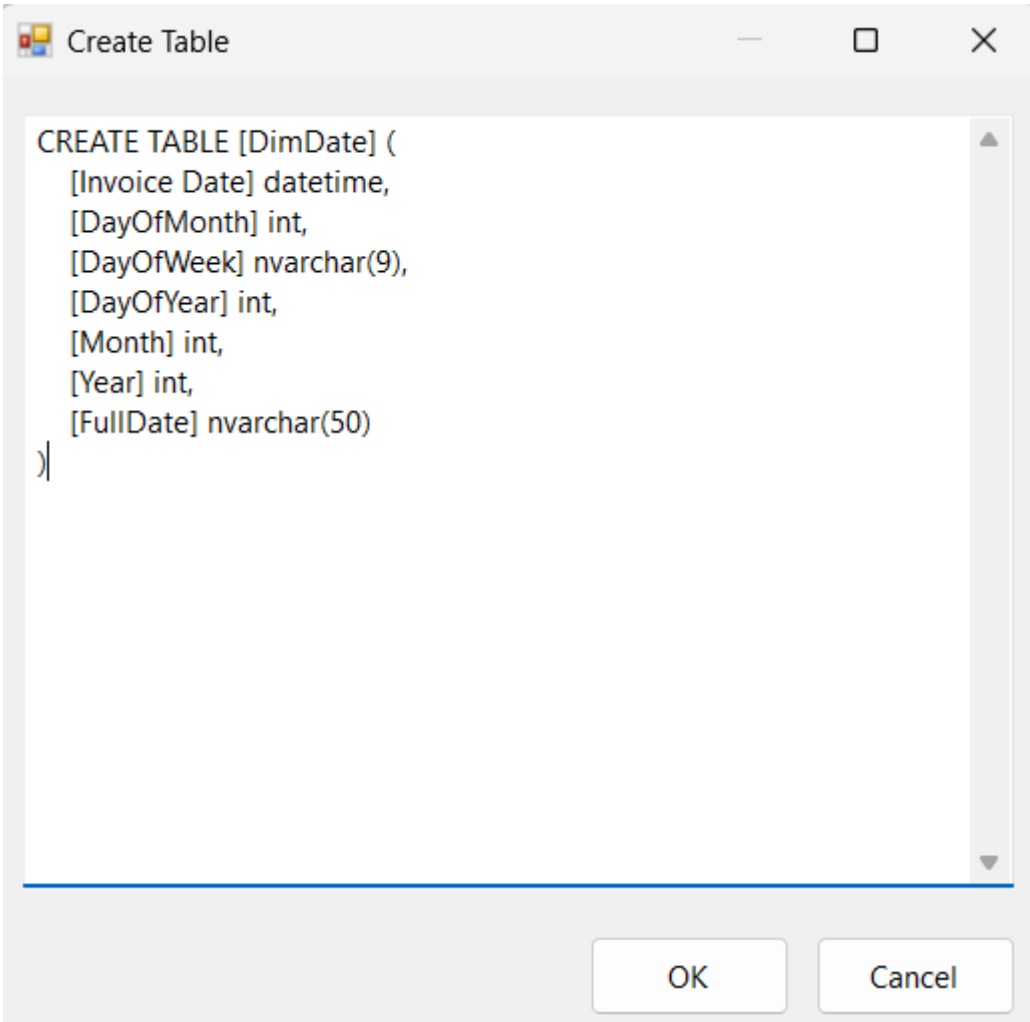
**DimDate:**

The data is read from the excel source and is sent to Sort tool in SSIS, it sorts out the data and then date is calculated in derived column, it comes out as a String which is then converted into Date data type and inserted into DimDate table.

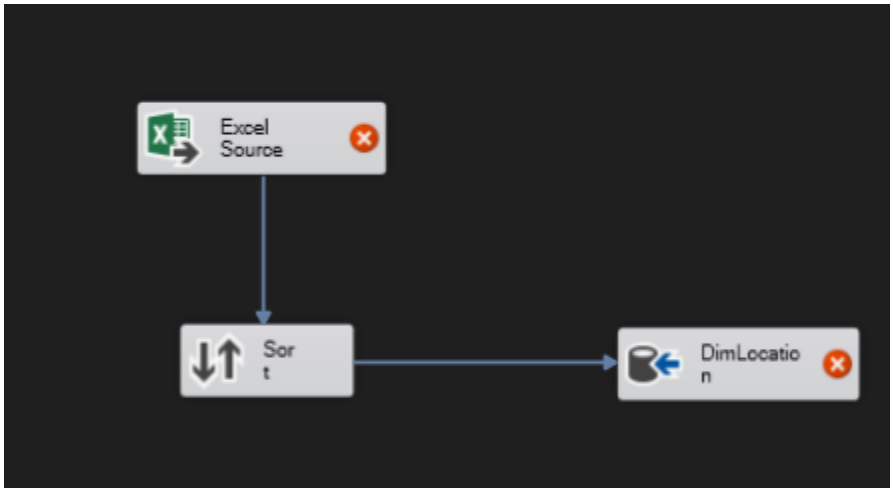



Derived Column Transformation Editor
Specify the expressions used to create new column values, and indicate whether the values update existing columns or populate new columns.
Variables and Parameters
Columns
Mathematical Functions
String Functions
Date/Time Functions
NULL Functions
Type Casts
Operators
Description:
Derived Column Name
Derived Column
Expression
Data Type
DayOfMonth
<add as new column>
DAY([Invoice Date])
four-byte signed inte...
DayOfWeek
<add as new column>
(DATEPART("dw",[Invoice Date]) == 1 ? "Sund...
Unicode string [DT_...
DayOfYear
<add as new column>
DATEPART("dy",[Invoice Date])
four-byte signed inte...
Month
<add as new column>
MONTH([Invoice Date])
four-byte signed inte...
Year
<add as new column>
YEAR([Invoice Date])
four-byte signed inte...
Configure Error Output...
OK
Cancel
Help

Below is the script to create DimDate table.




**DimLocation** was created using the SSIS package. Contains information relevant to the location-based information.

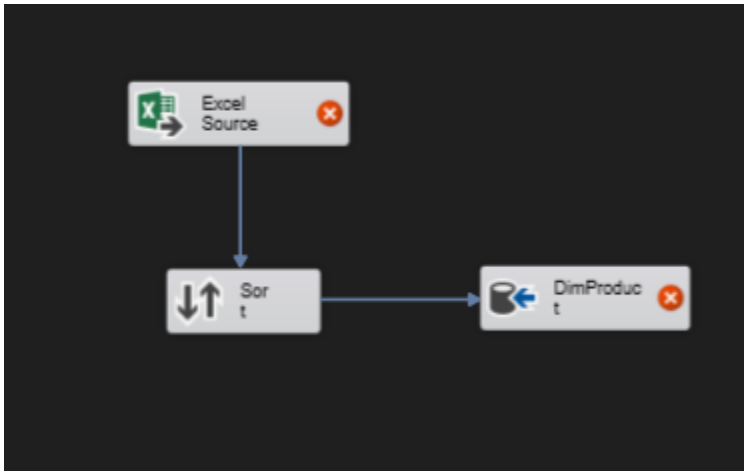



Create Table

```
CREATE TABLE [DimLocation] (
    [Region] nvarchar(255),
    [State] nvarchar(255),
    [City] nvarchar(255)
)
```

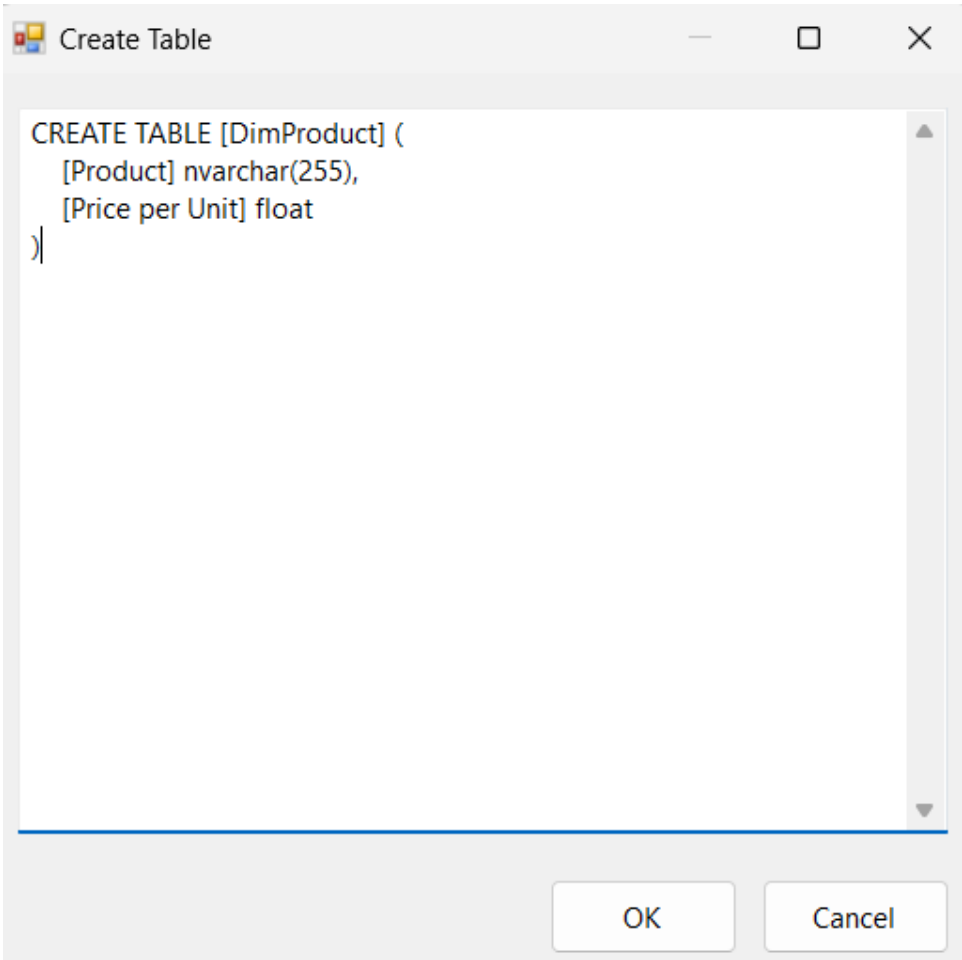
OK Cancel

**DimProduct** was created using the SSIS package. Contains information related to the product details.



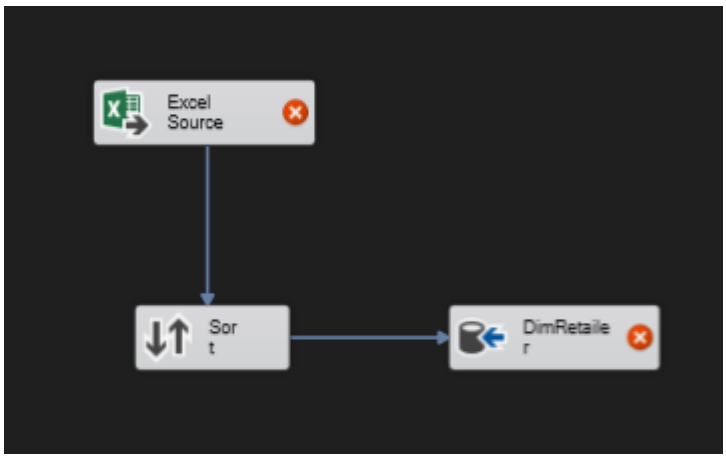

Data is read from Excel and is given to sort tool to do the sorting and then it is inserted in DimProduct table.

Below is the script of DimProduct table.

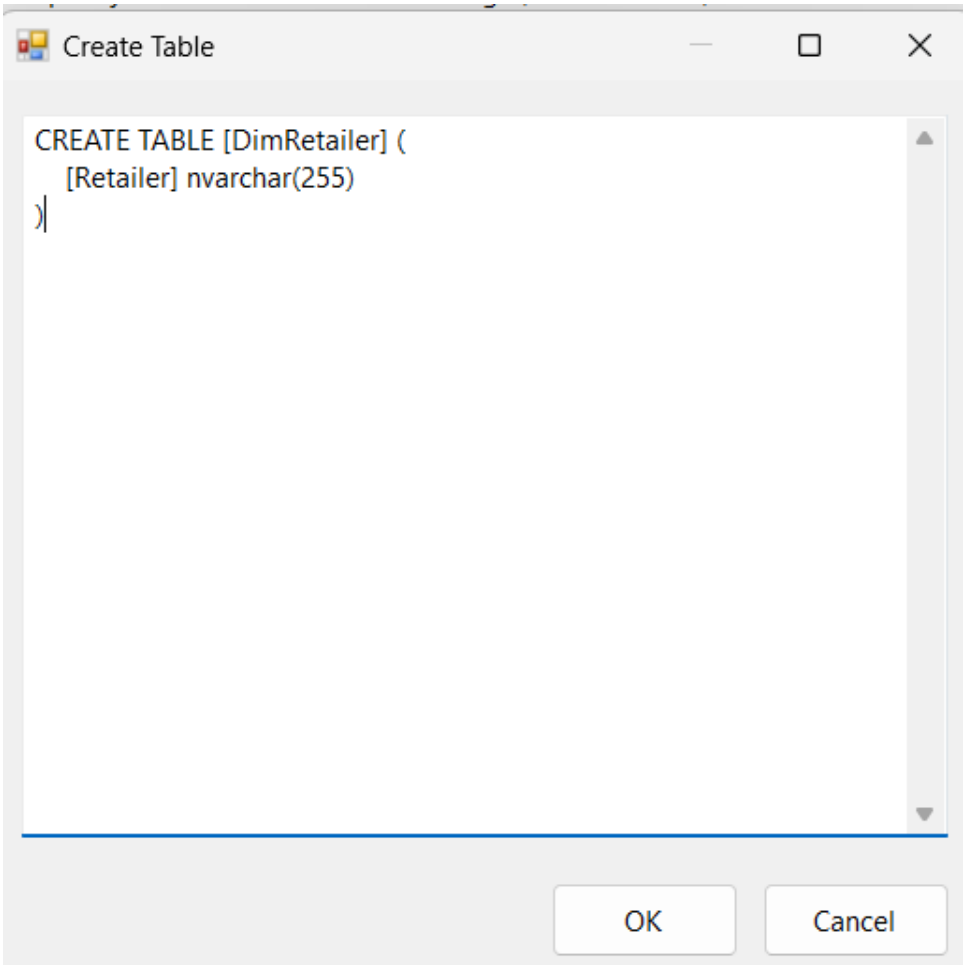



**DimRetailer** created using the SSIS package. Stores retailer related Information.

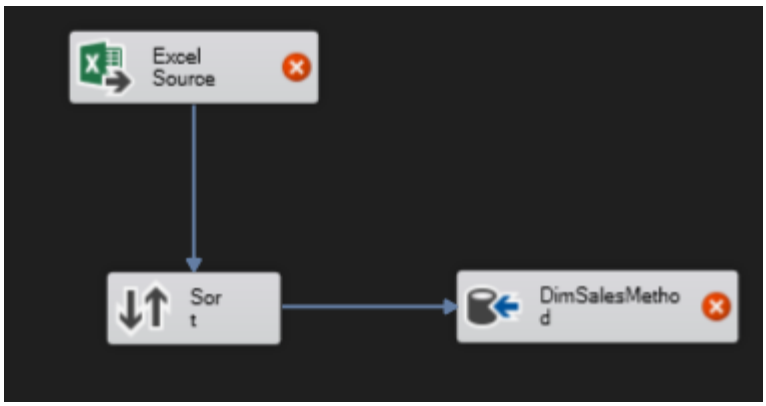
Data is read from the excel and is given to sort tool to do the sorting and then it is inserted in DimRetailer table.

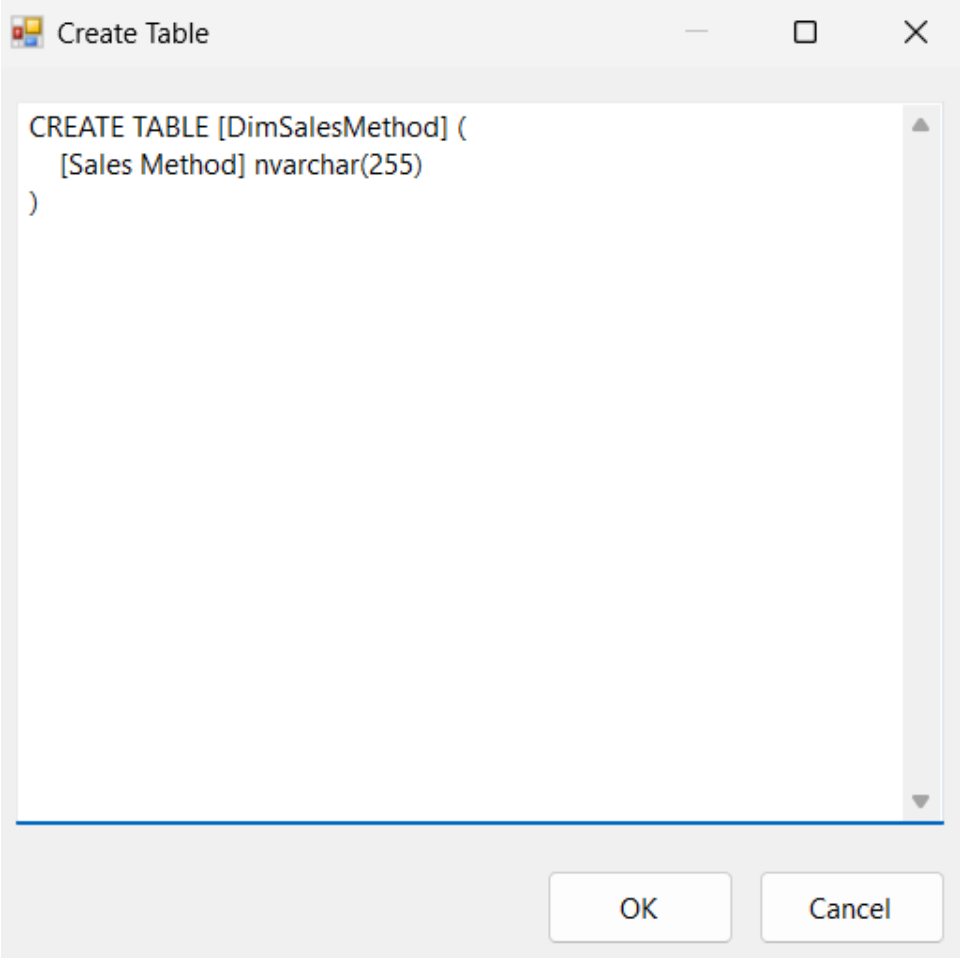


Below is the script of DimRetailer table.

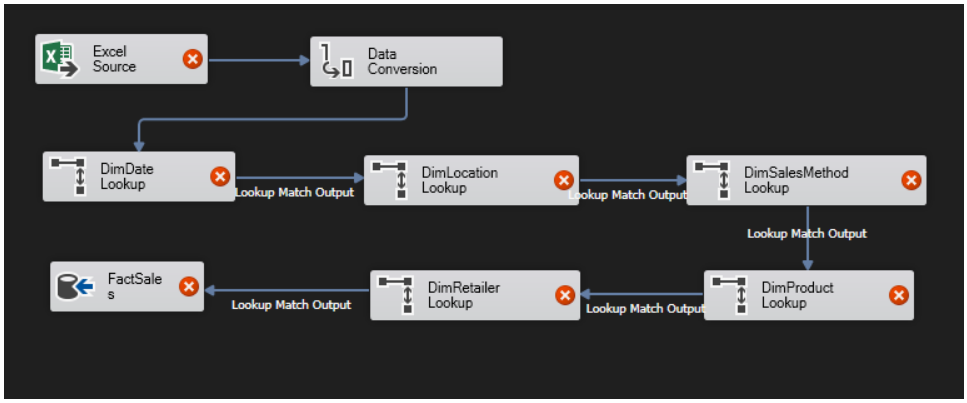


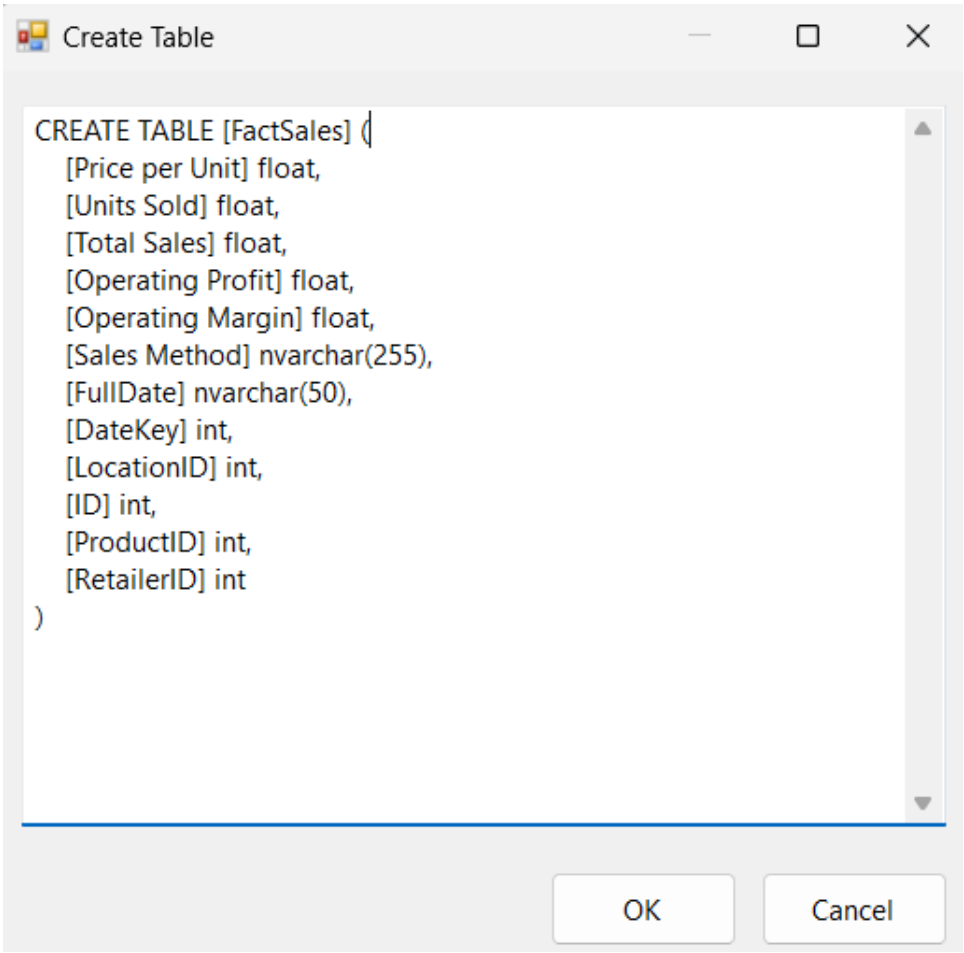

**DimSalesMethod** created using the SSIS package.






**FactSales** table created using the SSIS package and has data regarding the whole warehouse and the previously created dimensions table.



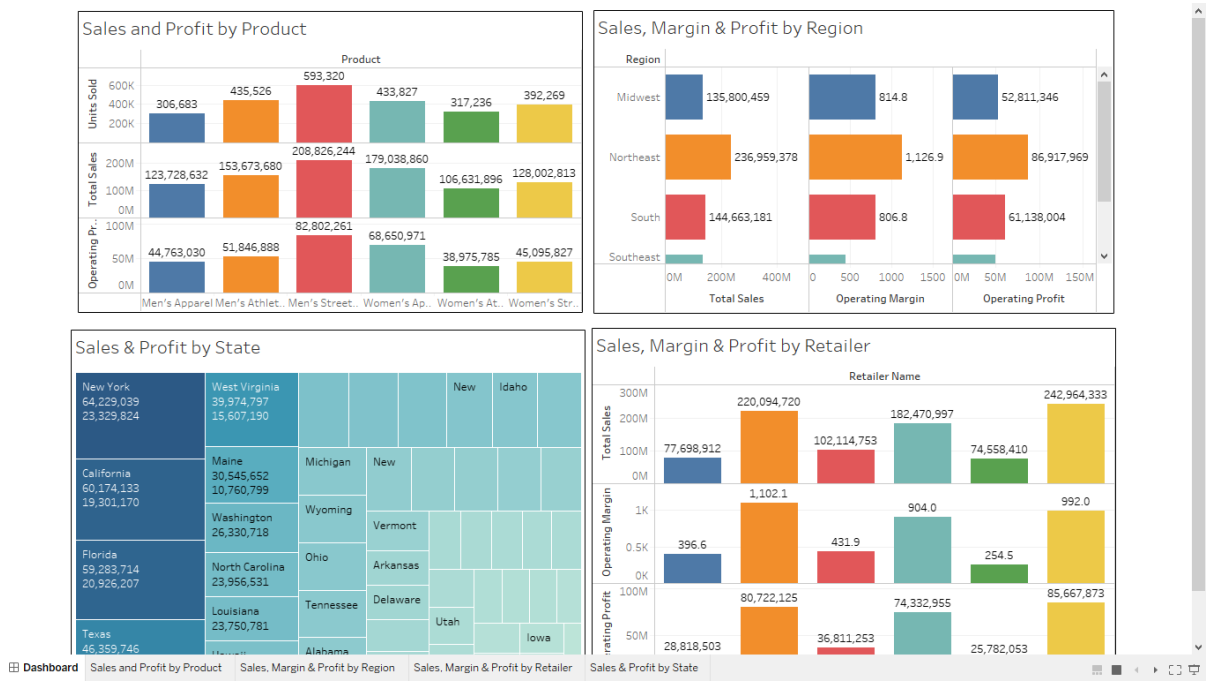



## 4. Visualizations and Reports:

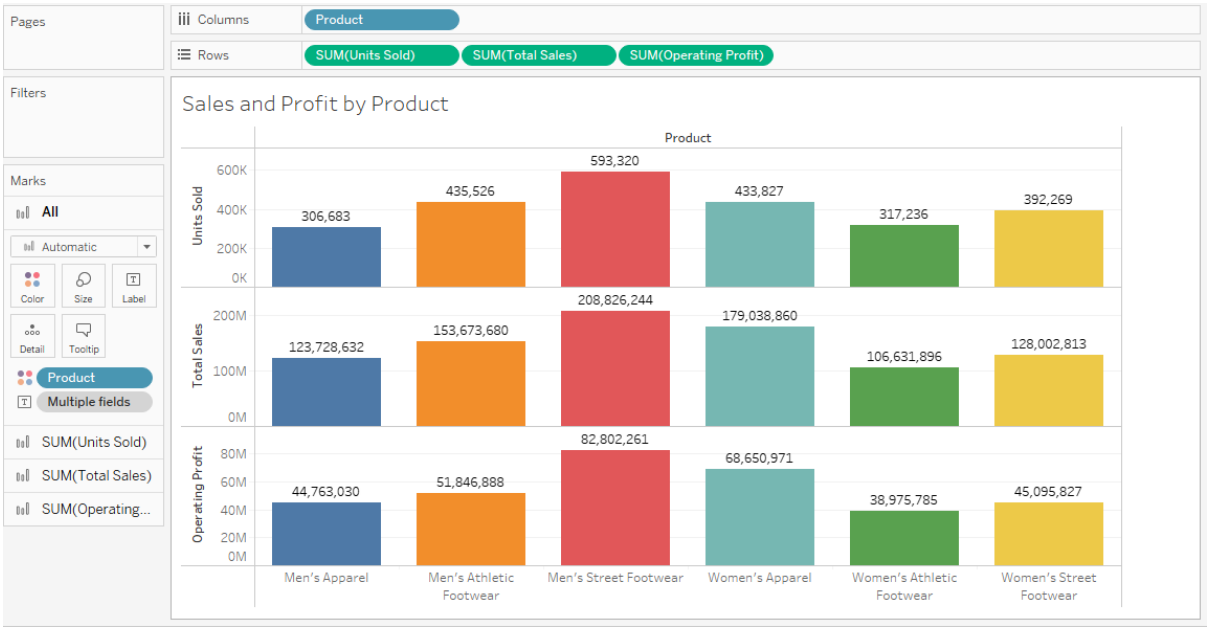
### 4.1. TABLEAU Visualizations:

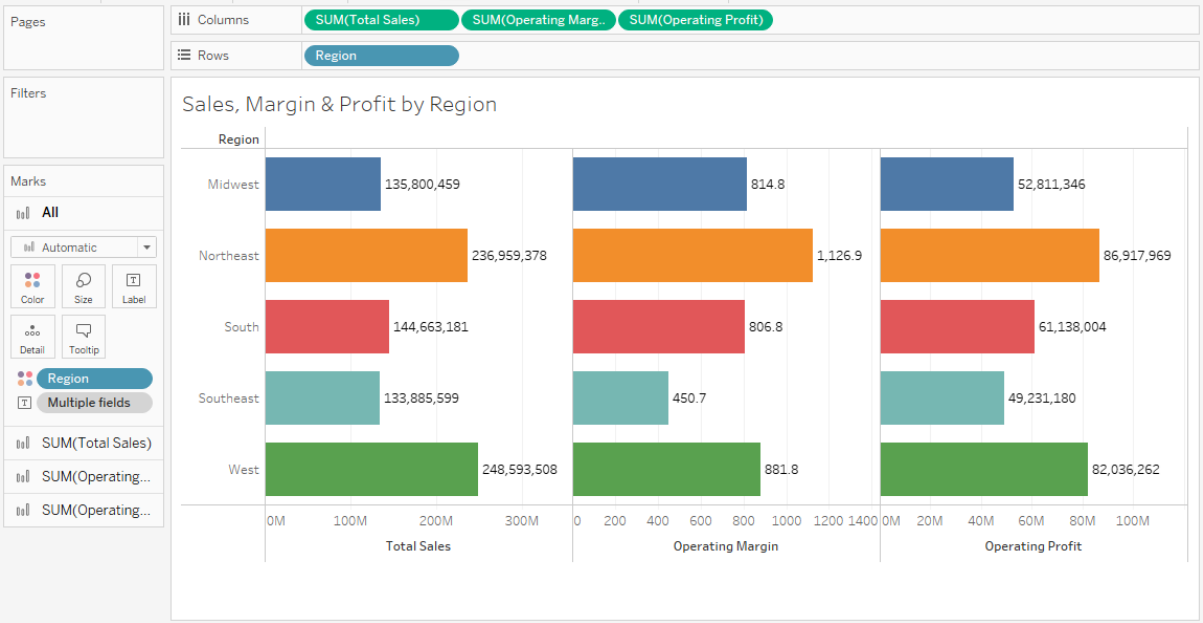
As per the business requirements, it is necessary to know the sales by product, profit by product, sales by region, margin by region and profit by region. By knowing this information, it can be easy for a company or a business to decide what measures should be taken in order to increase their sales or increase their margins thereby increasing the profits, hence it is done using below graphs in tableau.

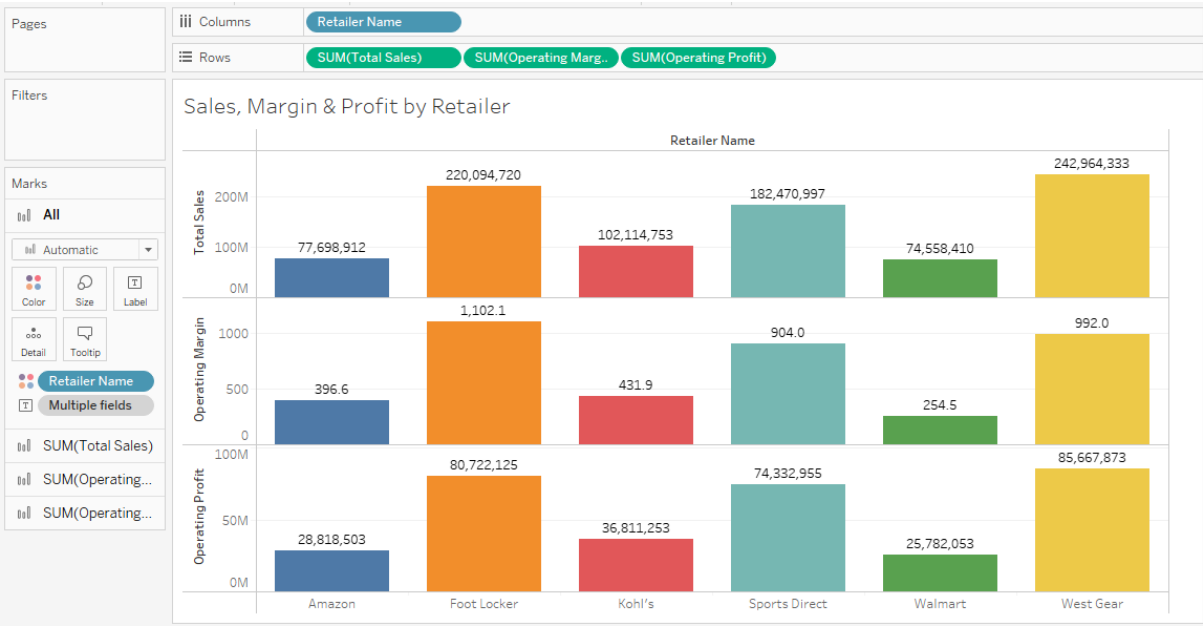
(Khedikar, 2021)



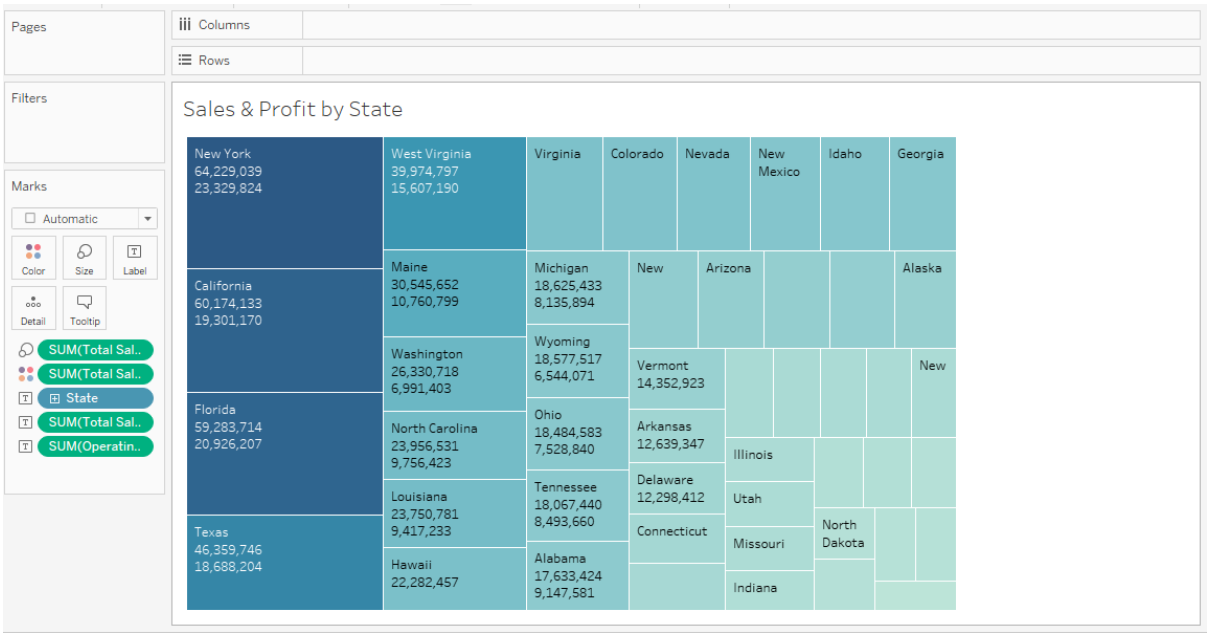

Another visualization is created to see sales and profit by product. The data clearly shows the information regarding units sold, total sales and operating profit of a business.



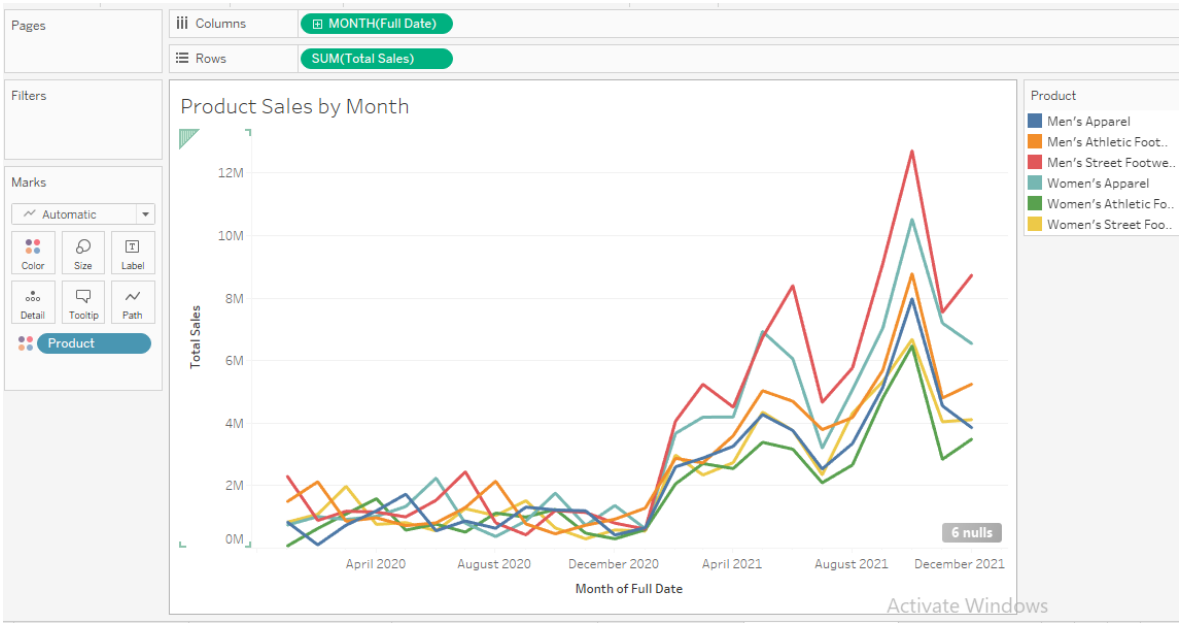


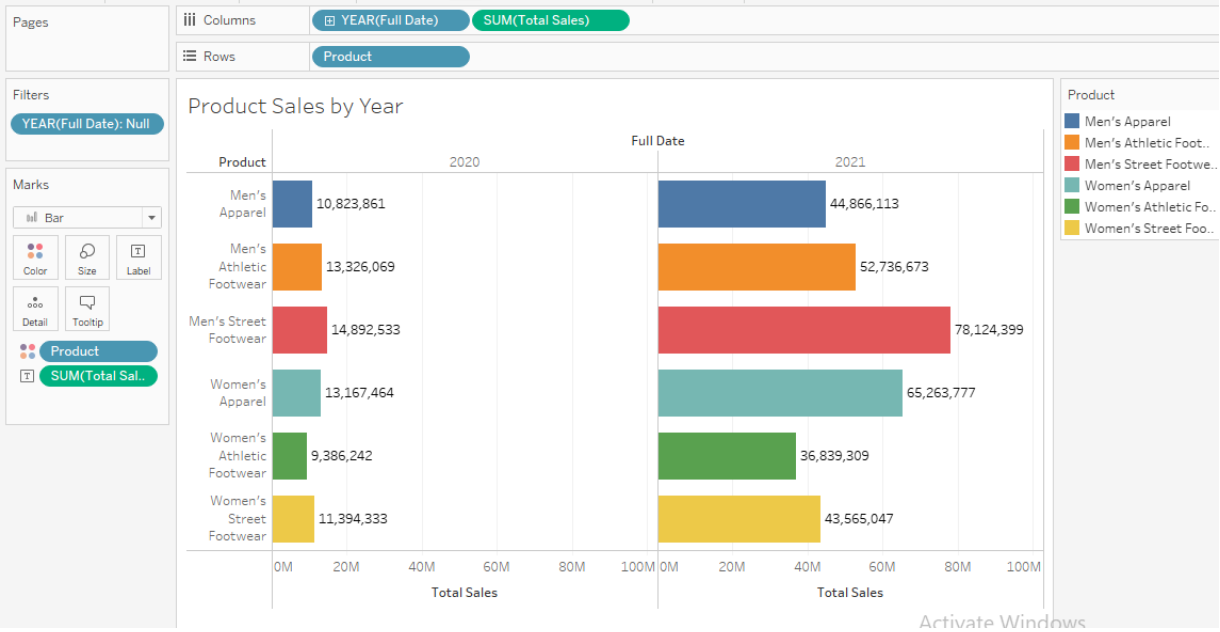




One more requirement of the business is to understand the sales and profit by state, below visualization clarifies that, it shows the sales and profit by state. This is an important business requirement because it tells which state is lagging in terms of volume of sales. It makes it easier to understand because we can look at the size of block and compare the states easily instead of working on numeric values.




Another important business requirement is to understand sales by gender, i.e. how many Men's apparels were sold and how many women's apparels were sold, below visualization mentions about that.






## 4.2. REPORTS - SSRS

Visualizations of the data warehouse using the SSRS reports. (Fana et al., 2021)

**Report 1:**

The below code shows the visualization and analysis of the sales by day of the week against the trend of the retailer values.

Query Designer

Edit as Text | Import... | Command type: Text

```
SELECT
   dd.DayOfWeek,
   dr.RetailerName,
   SUM(fs.TotalSales) AS TotalSales,
   AVG(SUM(fs.TotalSales)) OVER (PARTITION BY dd.DayOfWeek) AS AvgSales_DayOfWeek
FROM
   FactSales AS fs
JOIN
   DimDate AS dd ON fs.DateKey = dd.DateKey
JOIN
   DimRetailer AS dr ON fs.RetailerID = dr.RetailerID
GROUP BY
   dd.DayOfWeek,
   dr.RetailerName
ORDER BY
   dd.DayOfWeek,
   TotalSales DESC;
```

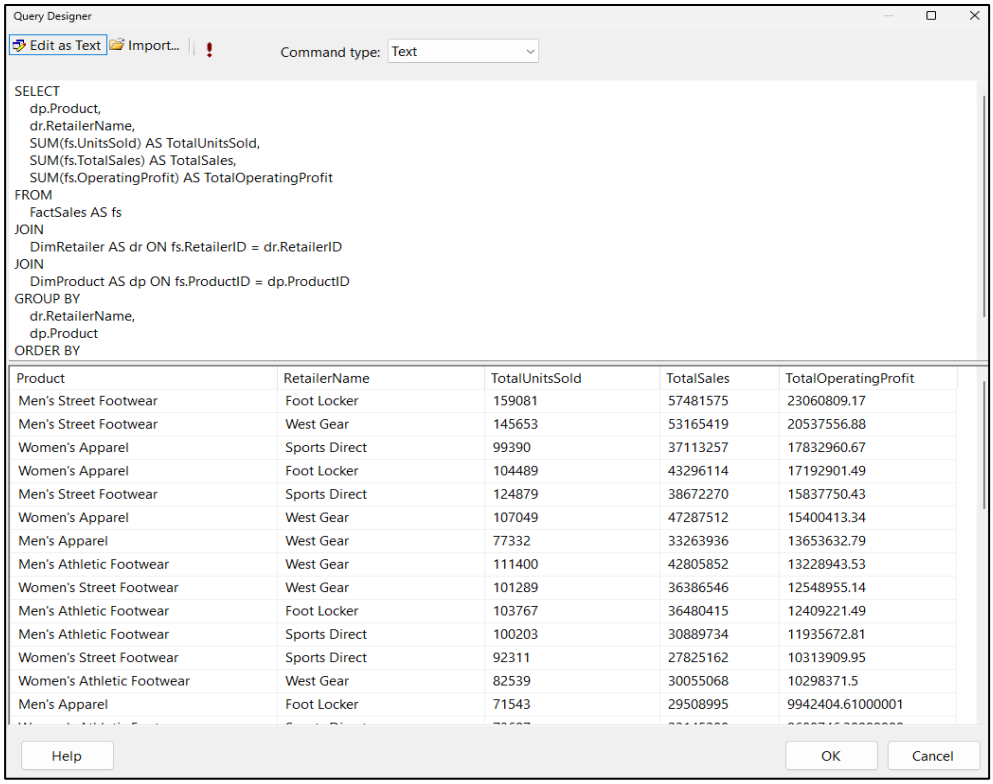
| DayOfWeek | RetailerName | TotalSales | AvgSales_DayOfWeek |
|---|---|---|---|
| Friday | West Gear | 40799041 | 24447183.1666667 |
| Friday | Foot Locker | 35760575 | 24447183.1666667 |
| Friday | Sports Direct | 27376997 | 24447183.1666667 |
| Friday | Amazon | 15207839 | 24447183.1666667 |
| Friday | Kohl's | 15182675 | 24447183.1666667 |
| Friday | Walmart | 12355972 | 24447183.1666667 |
| Monday | Foot Locker | 27710915 | 18722564.5 |
| Monday | Sports Direct | 25616521 | 18722564.5 |
| Monday | West Gear | 24488092 | 18722564.5 |
| Monday | Kohl's | 15788112 | 18722564.5 |
| Monday | Amazon | 10247198 | 18722564.5 |
| Monday | Walmart | 8484549 | 18722564.5 |
| Saturday | West Gear | 28904854 | 19654996.5 |

Help | OK | Cancel

## Sales by Day of Week and Retailer with Trend Analysis

| Day Of Week | Friday | | Monday | | Saturday | |
|---|---|---|---|---|---|---|
| | Total Sales | Avg Sales Day Of Week | Total Sales | Avg Sales Day Of Week | Total Sales | Avg Sales Day Of Week |
| Amazon | 15207839 | 24447183.1666667 | 10247198 | 18722564.5 | 10496401 | 19654996.5 |
| Foot Locker | 35760575 | 24447183.1666667 | 27710915 | 18722564.5 | 28473406 | 19654996.5 |
| Kohl's | 15182675 | 24447183.1666667 | 15788112 | 18722564.5 | 15215618 | 19654996.5 |
| Sports Direct | 27376997 | 24447183.1666667 | 25616521 | 18722564.5 | 24710781 | 19654996.5 |
| Walmart | 12355972 | 24447183.1666667 | 8484549 | 18722564.5 | 10128919 | 19654996.5 |
| West Gear | 40799041 | 24447183.1666667 | 24488092 | 18722564.5 | 28904854 | 19654996.5 |

**Report 2:** The below code shows the visualization and analysis of the sales by retailer and product by the operating profit.



Query Designer

Edit as Text | Import... | Command type: Text

```
SELECT
    dp.Product,
    dr.RetailerName,
    SUM(fs.UnitsSold) AS TotalUnitsSold,
    SUM(fs.TotalSales) AS TotalSales,
    SUM(fs.OperatingProfit) AS TotalOperatingProfit
FROM
    FactSales AS fs
JOIN
    DimRetailer AS dr ON fs.RetailerID = dr.RetailerID
JOIN
    DimProduct AS dp ON fs.ProductID = dp.ProductID
GROUP BY
    dr.RetailerName,
    dp.Product
ORDER BY
```

| Product | RetailerName | TotalUnitsSold | TotalSales | TotalOperatingProfit |
|---|---|---|---|---|
| Men's Street Footwear | Foot Locker | 159081 | 57481575 | 23060809.17 |
| Men's Street Footwear | West Gear | 145653 | 53165419 | 20537556.88 |
| Women's Apparel | Sports Direct | 99390 | 37113257 | 17832960.67 |
| Women's Apparel | Foot Locker | 104489 | 43296114 | 17192901.49 |
| Men's Street Footwear | Sports Direct | 124879 | 38672270 | 15837750.43 |
| Women's Apparel | West Gear | 107049 | 47287512 | 15400413.34 |
| Men's Apparel | West Gear | 77332 | 33263936 | 13653632.79 |
| Men's Athletic Footwear | West Gear | 111400 | 42805852 | 13228943.53 |
| Women's Street Footwear | West Gear | 101289 | 36386546 | 12548955.14 |
| Men's Athletic Footwear | Foot Locker | 103767 | 36480415 | 12409221.49 |
| Men's Athletic Footwear | Sports Direct | 100203 | 30889734 | 11935672.81 |
| Women's Street Footwear | Sports Direct | 92311 | 27825162 | 10313909.95 |
| Women's Athletic Footwear | West Gear | 82539 | 30055068 | 10298371.5 |
| Men's Apparel | Foot Locker | 71543 | 29508995 | 9942404.61000001 |

Help | OK | Cancel

## Sales by Retailer and Product with Operating Profit

| Product | Retailer Name | Total Units Sold | Total Sales | Total Operating Profit |
|---|---|---|---|---|
| Men's Apparel | West Gear | 77332 | 33263936 | 13653632.79 |
| | Foot Locker | 71543 | 29508995 | 9942404.61000001 |
| | Sports Direct | 67160 | 24825286 | 8723914.71 |
| | Kohl's | 36080 | 14164965 | 5945043.2 |
| | Amazon | 24062 | 10474770 | 3331443.8 |
| | Walmart | 30506 | 11490680 | 3166591.22 |
| Men's Athletic Footwear | West Gear | 111400 | 42805852 | 13228943.53 |
| | Foot Locker | 103767 | 36480415 | 12409221.49 |
| | Sports Direct | 100203 | 30889734 | 11935672.81 |
| | Kohl's | 51012 | 17885316 | 5725762.6 |
| | Amazon | 31987 | 12011959 | 4518030.11 |
| | Walmart | 37157 | 13600404 | 4029257.65 |
| Men's Street Footwear | Foot Locker | 159081 | 57481575 | 23060809.17 |
| | West Gear | 145653 | 53165419 | 20537556.88 |
| | Sports Direct | 124879 | 38672270 | 15837750.43 |
| | Kohl's | 66844 | 22421073 | 9219819.89000001 |
| | Amazon | 54883 | 22161652 | 8707658.12 |

**Report 3:** This report presents an analysis of sales by month and region with the operating

Query Designer

Edit as Text | Import... | Command type: Text

```
SELECT
    dl.Region,
    CONVERT(nvarchar(7), dd.FullDate, 120) AS Month,
    SUM(fs.TotalSales) AS TotalSales,
    SUM(fs.OperatingProfit) / NULLIF(SUM(fs.TotalSales), 0) AS OperatingMargin
FROM
    FactSales AS fs
JOIN
    DimDate AS dd ON fs.DateKey = dd.DateKey
JOIN
    DimLocation AS dl ON fs.LocationID = dl.LocationID
GROUP BY
    CONVERT(nvarchar(7), dd.FullDate, 120),
    dl.Region
ORDER BY
    Month ASC,
    OperatingMargin DESC;
```

| Region | Month | TotalSales | OperatingMargin |
|---|---|---|---|
| Midwest | 01-01-2 | 141119 | 0.603846753449217 |
| Northeast | 01-01-2 | 845141 | 0.452151581807059 |
| West | 01-01-2 | 513783 | 0.253545582473535 |
| West | 01-02-2 | 845019 | 0.382267120620957 |
| Northeast | 01-02-2 | 472023 | 0.303263675710717 |
| Midwest | 01-02-2 | 368250 | 0.252464358452138 |
| West | 01-03-2 | 237739 | 0.503225259633464 |
| Midwest | 01-03-2 | 174641 | 0.403986177358123 |
| South | 01-04-2 | 94452 | 0.35315715919197 |
| West | 01-04-2 | 197440 | 0.1535028363047 |
| South | 01-05-2 | 210320 | 0.403617344998098 |
| West | 01-05-2 | 800260 | 0.342021443031015 |

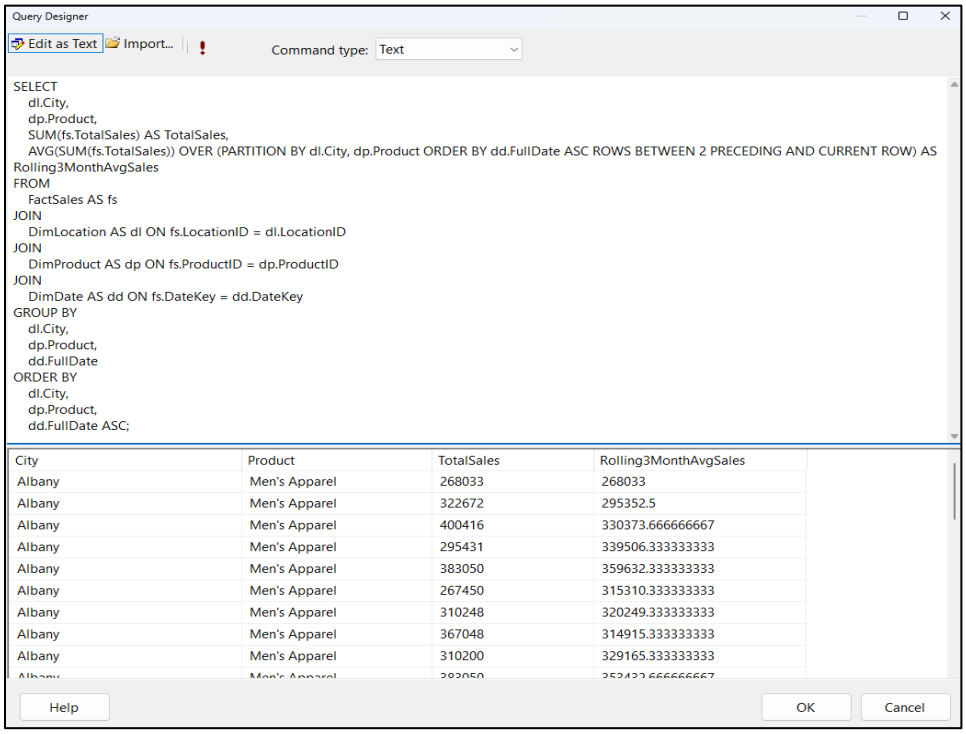
Cancel

## Sales by Month and Region with Operating Margin

| Region | Month | Total Sales | Operating Margin |
|---|---|---|---|
| ⊟ Midwest | | | |
| | 01-01-2 | 141119 | 0.603846753449217 |
| | 01-02-2 | 368250 | 0.252464358452138 |
| | 01-03-2 | 174641 | 0.403986177358123 |
| | 01-06-2 | 71920 | 0.503203559510567 |
| | 01-07-2 | 294668 | 0.44423968669825 |
| | 01-08-2 | 238709 | 0.382367862125014 |
| | 01-11-2 | 8974 | 0.353744149765991 |
| | 01-12-2 | 62801 | 0.353457588254964 |
| | 02-01-2 | 219684 | 0.25378161359043 |
| | 02-02-2 | 307511 | 0.453419519952132 |
| | 02-03-2 | 463709 | 0.471024435583523 |
| | 02-06-2 | 172636 | 0.403759586644732 |
| | 02-07-2 | 1211204 | 0.365663694967982 |
| | 02-08-2 | 298122 | 0.403732364602411 |
| | 02-11-2 | 38393 | 0.302791133800432 |
| | 02-12-2 | 1256538 | 0.353232277893705 |
| | 03-01-2 | 620271 | 0.387709871975314 |
| | 03-02-2 | 636851 | 0.331950189290745 |
| | 03-03-2 | 292448 | 0.253310810810811 |
| | 03-04-2 | 326844 | 0.38078924502209 |
| | 03-06-2 | 673752 | 0.359344150369869 |
| | 03-07-2 | 198896 | 0.371578513393934 |
| | 03-08-2 | 1108686 | 0.360936045011843 |
| | 03-10-2 | 529973 | 0.356688887924479 |

margin.

**Report 4:** The code is for showing comparison between sales by location and product category with rolling 3 month average.




| City | Product | TotalSales | Rolling3MonthAvgSales |
|---|---|---|---|
| Albany | Men's Apparel | 268033 | 268033 |
| Albany | Men's Apparel | 322672 | 295352.5 |
| Albany | Men's Apparel | 400416 | 330373.666666667 |
| Albany | Men's Apparel | 295431 | 339506.333333333 |
| Albany | Men's Apparel | 383050 | 359632.333333333 |
| Albany | Men's Apparel | 267450 | 315310.333333333 |
| Albany | Men's Apparel | 310248 | 320249.333333333 |
| Albany | Men's Apparel | 367048 | 314915.333333333 |
| Albany | Men's Apparel | 310200 | 329165.333333333 |
| Albany | Men's Apparel | 383050 | 353432.666666667 |

# Sales by Location and Product Category with Rolling 3-Month Average

| City | Product | Total Sales | Rolling3Month Avg Sales |
|---|---|---|---|
| **Albany** | | **24427804** | **24427804** |
| | **Men's Apparel** | 3873553 | 3873553 |
| | **Men's Athletic Footwear** | 3829596 | 3829596 |
| | **Men's Street Footwear** | 5773987 | 5773987 |
| | **Women's Apparel** | 4779109 | 4779109 |
| | **Women's Athletic Footwear** | 2973011 | 2973011 |
| | **Women's Street Footwear** | 3198548 | 3198548 |
| **Albuquerque** | | **19865016** | **19865016** |
| | **Men's Apparel** | 2528344 | 2528344 |
| | **Men's Athletic Footwear** | 3803130 | 3803130 |
| | **Men's Street Footwear** | 3892153 | 3892153 |
| | **Women's Apparel** | 3868520 | 3868520 |
| | **Women's Athletic Footwear** | 2582792 | 2582792 |

## 5. CONCLUSIONS:

In conclusion, in this project I learnt to create a data warehouse to organize data effectively for analysis. I also learned how to ensure that data is accurate and complete with the help of SQL Server and SSIS. The Business Intelligence reports that I developed with SSRS gave me valuable insights into different areas that help to make better decisions. I further learnt to visualize and explore data in a more interactive and meaningful way using Tableau. Overall, through this project helped me gain business intelligence skills.

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## APPENDIX A – CREATE TABLE QUERIES FOR DATA WAREHOUSE / DATA MART:

```
USE [master]

GO

/****** Object:  Database [AdidasDB]    Script Date: 26-03-2024 11.23.38 PM ******/

CREATE DATABASE [AdidasDB]

 CONTAINMENT = NONE

ON  PRIMARY

( NAME = N'AdidasDB', FILENAME = N'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\AdidasDB.mdf' , SIZE = 8192KB , MAXSIZE = UNLIMITED, FILEGROWTH = 65536KB )
```

```
 LOG ON
( NAME = N'AdidasDB_log', FILENAME = N'C:\Program Files\Microsoft SQL Server\MSSQL16.SQLEXPRESS\MSSQL\DATA\AdidasDB_log.ldf' , SIZE = 73728KB , MAXSIZE = 2048GB , FILEGROWTH = 65536KB )
 WITH CATALOG_COLLATION = DATABASE_DEFAULT, LEDGER = OFF
GO
ALTER DATABASE [AdidasDB] SET COMPATIBILITY_LEVEL = 160
GO
IF (1 = FULLTEXTSERVICEPROPERTY('IsFullTextInstalled'))
begin
EXEC [AdidasDB].[dbo].[sp_fulltext_database] @action = 'enable'
end
GO
ALTER DATABASE [AdidasDB] SET ANSI_NULL_DEFAULT OFF
GO
ALTER DATABASE [AdidasDB] SET ANSI_NULLS OFF
GO
ALTER DATABASE [AdidasDB] SET ANSI_PADDING OFF
GO
ALTER DATABASE [AdidasDB] SET ANSI_WARNINGS OFF
GO
ALTER DATABASE [AdidasDB] SET ARITHABORT OFF
GO
ALTER DATABASE [AdidasDB] SET AUTO_CLOSE OFF
GO
```

```
ALTER DATABASE [AdidasDB] SET AUTO_SHRINK OFF
GO
ALTER DATABASE [AdidasDB] SET AUTO_UPDATE_STATISTICS ON
GO
ALTER DATABASE [AdidasDB] SET CURSOR_CLOSE_ON_COMMIT OFF
GO
ALTER DATABASE [AdidasDB] SET CURSOR_DEFAULT  GLOBAL
GO
ALTER DATABASE [AdidasDB] SET CONCAT_NULL_YIELDS_NULL OFF
GO
ALTER DATABASE [AdidasDB] SET NUMERIC_ROUNDABORT OFF
GO
ALTER DATABASE [AdidasDB] SET QUOTED_IDENTIFIER OFF
GO
ALTER DATABASE [AdidasDB] SET RECURSIVE_TRIGGERS OFF
GO
ALTER DATABASE [AdidasDB] SET  DISABLE_BROKER
GO
ALTER DATABASE [AdidasDB] SET AUTO_UPDATE_STATISTICS_ASYNC OFF
GO
ALTER DATABASE [AdidasDB] SET DATE_CORRELATION_OPTIMIZATION OFF
GO
ALTER DATABASE [AdidasDB] SET TRUSTWORTHY OFF
GO
```

```
ALTER DATABASE [AdidasDB] SET ALLOW_SNAPSHOT_ISOLATION OFF
GO
ALTER DATABASE [AdidasDB] SET PARAMETERIZATION SIMPLE
GO
ALTER DATABASE [AdidasDB] SET READ_COMMITTED_SNAPSHOT OFF
GO
ALTER DATABASE [AdidasDB] SET HONOR_BROKER_PRIORITY OFF
GO
ALTER DATABASE [AdidasDB] SET RECOVERY SIMPLE
GO
ALTER DATABASE [AdidasDB] SET  MULTI_USER
GO
ALTER DATABASE [AdidasDB] SET PAGE_VERIFY CHECKSUM
GO
ALTER DATABASE [AdidasDB] SET DB_CHAINING OFF
GO
ALTER DATABASE [AdidasDB] SET FILESTREAM( NON_TRANSACTED_ACCESS = OFF
)
GO
ALTER DATABASE [AdidasDB] SET TARGET_RECOVERY_TIME = 60 SECONDS
GO
ALTER DATABASE [AdidasDB] SET DELAYED_DURABILITY = DISABLED
GO
ALTER DATABASE [AdidasDB] SET ACCELERATED_DATABASE_RECOVERY = OFF
GO
```

```
ALTER DATABASE [AdidasDB] SET QUERY_STORE = ON

GO

ALTER DATABASE [AdidasDB] SET QUERY_STORE (OPERATION_MODE =
READ_WRITE, CLEANUP_POLICY = (STALE_QUERY_THRESHOLD_DAYS = 30),
DATA_FLUSH_INTERVAL_SECONDS = 900, INTERVAL_LENGTH_MINUTES = 60,
MAX_STORAGE_SIZE_MB = 1000, QUERY_CAPTURE_MODE = AUTO,
SIZE_BASED_CLEANUP_MODE = AUTO, MAX_PLANS_PER_QUERY = 200,
WAIT_STATS_CAPTURE_MODE = ON)

GO

USE [AdidasDB]

GO

/****** Object:  Table [dbo].[DimDate]    Script Date: 26-03-2024 11.23.38 PM ******/

SET ANSI_NULLS ON

GO

SET QUOTED_IDENTIFIER ON

GO

CREATE TABLE [dbo].[DimDate](

        [DateKey] [int] IDENTITY(1,1) NOT NULL,

        [FullDate] [nvarchar](50) NULL,

        [DayOfMonth] [int] NULL,

        [DayOfWeek] [nvarchar](9) NULL,

        [DayOfYear] [int] NULL,

        [Month] [int] NULL,

        [Year] [int] NULL,

PRIMARY KEY CLUSTERED

(
```

```
	[DateKey] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

) ON [PRIMARY]

GO

/****** Object:  Table [dbo].[DimLocation]    Script Date: 26-03-2024 11.23.38 PM ******/

SET ANSI_NULLS ON

GO

SET QUOTED_IDENTIFIER ON

GO

CREATE TABLE [dbo].[DimLocation](

	[LocationID] [int] IDENTITY(1,1) NOT NULL,

	[Region] [nvarchar](255) NULL,

	[State] [nvarchar](255) NULL,

	[City] [nvarchar](255) NULL,

PRIMARY KEY CLUSTERED

(

	[LocationID] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

) ON [PRIMARY]

GO

/****** Object:  Table [dbo].[DimProduct]    Script Date: 26-03-2024 11.23.38 PM ******/

SET ANSI_NULLS ON
```

```
GO

SET QUOTED_IDENTIFIER ON

GO

CREATE TABLE [dbo].[DimProduct](

	[ProductID] [int] IDENTITY(1,1) NOT NULL,

	[Product] [nvarchar](255) NULL,

	[PricePerUnit] [float] NULL,

PRIMARY KEY CLUSTERED

(

	[ProductID] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

) ON [PRIMARY]

GO

/****** Object:  Table [dbo].[DimRetailer]    Script Date: 26-03-2024 11.23.38 PM ******/

SET ANSI_NULLS ON

GO

SET QUOTED_IDENTIFIER ON

GO

CREATE TABLE [dbo].[DimRetailer](

	[RetailerID] [int] IDENTITY(1,1) NOT NULL,

	[RetailerName] [nvarchar](255) NULL,

PRIMARY KEY CLUSTERED

(
```

```
	[RetailerID] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

) ON [PRIMARY]

GO

/****** Object:  Table [dbo].[DimSalesMethod]    Script Date: 26-03-2024 11.23.38 PM ******/

SET ANSI_NULLS ON

GO

SET QUOTED_IDENTIFIER ON

GO

CREATE TABLE [dbo].[DimSalesMethod](

	[ID] [int] IDENTITY(1,1) NOT NULL,

	[SalesMethod] [nvarchar](255) NULL,

PRIMARY KEY CLUSTERED

(

	[ID] ASC

)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]

) ON [PRIMARY]

GO

/****** Object:  Table [dbo].[FactSales]    Script Date: 26-03-2024 11.23.38 PM ******/

SET ANSI_NULLS ON

GO

SET QUOTED_IDENTIFIER ON
```

```
GO
CREATE TABLE [dbo].[FactSales](
	[SalesID] [int] IDENTITY(1,1) NOT NULL,
	[DateKey] [int] NULL,
	[LocationID] [int] NULL,
	[SalesMethodID] [int] NULL,
	[ProductID] [int] NULL,
	[RetailerID] [int] NULL,
	[UnitsSold] [float] NULL,
	[TotalSales] [float] NULL,
	[OperatingProfit] [float] NULL,
	[OperatingMargin] [float] NULL,
PRIMARY KEY CLUSTERED
(
	[SalesID] ASC
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF, IGNORE_DUP_KEY =
OFF, ALLOW_ROW_LOCKS = ON, ALLOW_PAGE_LOCKS = ON,
OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
USE [master]
GO
ALTER DATABASE [AdidasDB] SET  READ_WRITE
GO
```

## APPENDIX B – SSRS REPORT QUERIES:

**QUERY 1:**

```
SELECT

dd.DayOfWeek,

dr.RetailerName,

SUM (fs.TotalSales) AS TotalSales,

   AVG(SUM (fs.TotalSales)) OVER (PARTITION BY dd.DayOfWeek) AS
AvgSales_DayOfWeek

FROM

FactSales AS fs

JOIN

DimDate AS dd ON fs.DateKey = dd.DateKey

JOIN

DimRetailer AS dr ON fs.RetailerID = dr.RetailerID

GROUP BY

dd.DayOfWeek,

dr.RetailerName

ORDER BY

dd.DayOfWeek,

TotalSales DESC.
```

**QUERY 2:**

```
SELECT

dp.Product,

dr.RetailerName,
```

```
SUM (fs.UnitsSold) AS TotalUnitsSold,
SUM (fs.TotalSales) AS TotalSales,
SUM (fs.OperatingProfit) AS TotalOperatingProfit
FROM
FactSales AS fs
JOIN
DimRetailer AS dr ON fs.RetailerID = dr.RetailerID
JOIN
DimProduct AS dp ON fs.ProductID = dp.ProductID
GROUP BY
dr.RetailerName,
dp.Product
ORDER BY
TotalOperatingProfit DESC,
TotalSales DESC.
```

**QUERY 3:**

```
SELECT
dl.Region,
CONVERT (nvarchar(7), dd.FullDate, 120) AS Month,
SUM (fs.TotalSales) AS TotalSales,
SUM (fs.OperatingProfit) / NULLIF(SUM(fs.TotalSales), 0) AS OperatingMargin
FROM
FactSales AS fs
JOIN
```

```
DimDate AS dd ON fs.DateKey = dd.DateKey

JOIN

DimLocation AS dl ON fs.LocationID = dl.LocationID

GROUP BY

CONVERT (nvarchar(7), dd.FullDate, 120),

dl.Region

ORDER BY

   Month ASC,

OperatingMargin DESC.
```

**QUERY 4:**

```
SELECT

dl.City,

dp.Product,

SUM (fs.TotalSales) AS TotalSales,

   AVG(SUM (fs.TotalSales)) OVER (PARTITION BY dl.City, dp.Product ORDER BY
dd.FullDate ASC ROWS BETWEEN 2 PRECEDING AND CURRENT ROW) AS
Rolling3MonthAvgSales

FROM

FactSales AS fs

JOIN

DimLocation AS dl ON fs.LocationID = dl.LocationID

JOIN

DimProduct AS dp ON fs.ProductID = dp.ProductID

JOIN

DimDate AS dd ON fs.DateKey = dd.DateKey
```

```
GROUP BY
dl.City,
dp.Product,
dd.FullDate
ORDER BY
dl.City,
dp.Product,
dd.FullDate ASC;
```